

Assignment Report

Student: Denisenko Sofiia 狄苏菲

Student id: 1820249051

Course: Computer Organization and Architecture

Assignment №1

1.

Design a new processor for which the clock runs at 200 MHz. The following table gives instruction frequencies for Benchmark B. For this problem, we assume that (unlike many of today's computers) the processor only executes one instruction at a time.

Instruction Type	Frequency	Cycles
Loads & Stores	30%	6 cycles
Arithmetic Instructions	50%	4 cycles
All Others	20%	3 cycles

Clock rate = 200 MHz, 1 instruction/time

(1) Calculate the CPI for Benchmark B.

If we say that there are 100 instructions, then:

- 30 of them will be loads and stores.
- 50 of them will be arithmetic instructions.
- 20 of them will be branch instructions.

Clock cycles per 100 instructions are

$$(30 * 6) + (50 * 4) + (20 * 3) = 440 \text{ cycles/100 instructions}$$

Therefore, there are **4.4 Cycles** per instruction (CPI).

(2) The CPU execution time on the benchmark is exactly 11 seconds. What is the MIPS?

$$\text{MIPS} = \text{Clock rate} / (\text{CPI} * 10^6)$$

$$\text{MIPS} = (200 * 10^6) / (4.4 * 10^6) = \mathbf{45,45}$$

(3) The hardware expert says that if you double the number of registers, the cycle time must be increased by 20%. What would the new clock speed be (in MHz)?

$$\text{CPU} = \text{IC} * \text{CPI} * \text{Cycle time} \Rightarrow \text{CPU}_2 = \text{CPU}_1 * 1,2$$

$$\text{Clock rate}_2 = \text{Clock rate}_1 / 1,2 = 200 / 1,2 = \mathbf{166,67}$$

(4) The compiler expert says that if you double the number of registers, then the compiler will generate code that requires only half the number of Loads & Stores. What would the new CPI be?

$$\text{CPI} = (15 * 6) + (50 * 4) + (20 * 3) = 350 \text{ cycles} / 100 \text{ instructions}$$

Therefore, there are **3.5 Cycles** per instruction (CPI).

(5) How many CPU seconds will the benchmark take if we double the number of registers (taking into account both changes described above)?

$$\text{CPU} = (\text{IC} * \text{CPI}) / \text{Clock rate} = 100 * 3.5 * 10^6 / 166,67 * 10^6 = \mathbf{2,1 \text{ seconds}}$$

2.

2. Assume the word length is 8 bits, what is the original code, inverse code and complement code of the flowing binary numbers? For decimal, use the fixed point representation.

Number	0	-0	0.1000	-0.1000
Original code	00000000	10000000	00000010	10000010
Inverse code	00000000	11111111	00000010	11111101
Complement code	00000000	11000000	00000010	11111110
Number	0.1111	-0.1111	-1101	1101
Original code	00000111	10000111	10110011	00001101
Inverse code	00000111	11111000	01001100	00001101
Complement code	00000111	11111001	01001101	00001101

3.

Assume the complement code of X is $1.A_1A_2A_3A_4A_5A_6$,

(1). If $X > -\frac{1}{2}$, what condition should $A_1A_2A_3A_4A_5A_6$ satisfy?

$$-1/2 = 1.000000$$

So, the A_1 should be zero (0). And other A_{2-6} can be 1 or 0.

(2). If $-\frac{1}{8} \geq X \geq -\frac{1}{4}$, what condition should $A_1A_2A_3A_4A_5A_6$ satisfy?

$$-1/8 = 1.001000$$

$$-1/4 = 1.110000$$

$A_{1-5} = 0$; A_6 could be 1 or 0

4. For a 16 bits binary code, what is the representation range for the following cases?

(1) Unsigned integer

Range: 0 to 65,535

(2) Signed integer by the original code

Range: -32,767 to 32,767.

(3) Signed integer by the complement code

Range: -32,768 to 32,767.

(4) Fixed point decimal by the original code

Range: 0 to 255.99609375

(5) Fixed point decimal by the complement code

Range: -128 to 127.99609375

5. What is IEEE-754 single-precision code of the following numbers?
-0.625, -1000.5, 624

-0.625

Sign bit: 1

Binary: 0.101

Normalized: 1.01×2^{-1}

Exponent (offset binary): $-1 + 127 = 126 = 01111110$

Fraction: 010000000000000000000000

1 0111 1110 0100 0000 0000 0000 0000 000 = **BF20000H**

-1000.5

Sign bit: 1

Binary: 1111101000.1

Normalized: 1.1111010001×2^9

Exponent (offset binary): $9 + 127 = 136 = 10001000$

Fraction: 111101000100000000000000

1 1000 1000 1111 0100 0100 0000 0000 000 = **C47A2000H**

624

Sign bit: 0

Binary: 1001110000

Normalized: 1.001110000×2^9

Exponent (offset binary): $9 + 127 = 136 = 10001000$

Fraction: 001110000000000000000000

0 1000 1000 0011 1000 0000 0000 0000 000 = **441C0000H**

6. Assume using odd parity both horizontally and vertically for the following data block, how many errors here?

A7	A6	A5	A4	A3	A2	A1	A0	Parity
1	0	0	1	1	0	1	1	0
0	0	1	1	0	1	0	1	1
1	1	0	1	0	0	0	0	0
1	1	1	0	0	0	0	0	0
0	1	0	0	1	1	1	1	0
Parity 1	0	1	0	1	1	1	1	

6.

In the first and last column parity should be 0 (because there're 3 1s -> odd number). So, **2 errors**.

7. What is the Hamming code of information 01101110?

Step 1: Calculation of the number of redundant bits.

$$n=8, 2^P \geq n + P + 1, p=4$$

Step 2: Positioning the redundant bits. (Location of P_i is 2^{i-1})

H	12	11	10	9	8	7	6	5	4	3	2	1
P					P4				P3		P2	P1
D	D8	D7	D6	D5		D4	D3	D2		D1		

Step 3: Assign value to the redundant digit.

Hamming Code	Index	Checking Group
H1(P1)	1	—
H2(P2)	2	—
H3(D1)	3=1+2	P1,P2
H4(P3)	4	—
H5(D2)	5=1+4	P1,P3
H6(D3)	6=2+4	P2,P3
H7(D4)	7=1+2+4	P1,P2,P3
H8(P4)	8	—
H9(D5)	9=1+8	P1,P4
H10(D6)	10=2+8	P2,P4
H11(D7)	11=1+2+8	P1,P2,P4
H12(D8)	12=4+8	P3,P4

Step 4: Compute the redundant digit (data 01101110).

$$P1 = D1 \oplus D2 \oplus D4 \oplus D5 \oplus D7 = 0 \oplus 1 \oplus 1 \oplus 0 \oplus 1 = 1$$

$$P2 = D1 \oplus D3 \oplus D4 \oplus D6 \oplus D7 = 0 \oplus 1 \oplus 1 \oplus 1 \oplus 1 = 0$$

$$P3 = D2 \oplus D3 \oplus D4 \oplus D8 = 1 \oplus 1 \oplus 1 \oplus 0 = 1$$

$$P4 = D5 \oplus D6 \oplus D7 \oplus D8 = 0 \oplus 1 \oplus 1 \oplus 0 = 0$$

The resulting **Hamming code 0110 0111 1001**.